

AI breaks rules with total confidence.

I built a way to **catch it — and fix it** — using
an idea borrowed from physics.

Swipe →

THE PROBLEM

Trained to sound right, not to be right.

Neural networks optimise for what **looks** plausible. Nothing inside them forbids a **logically impossible** answer — an invalid configuration, a self-contradiction.

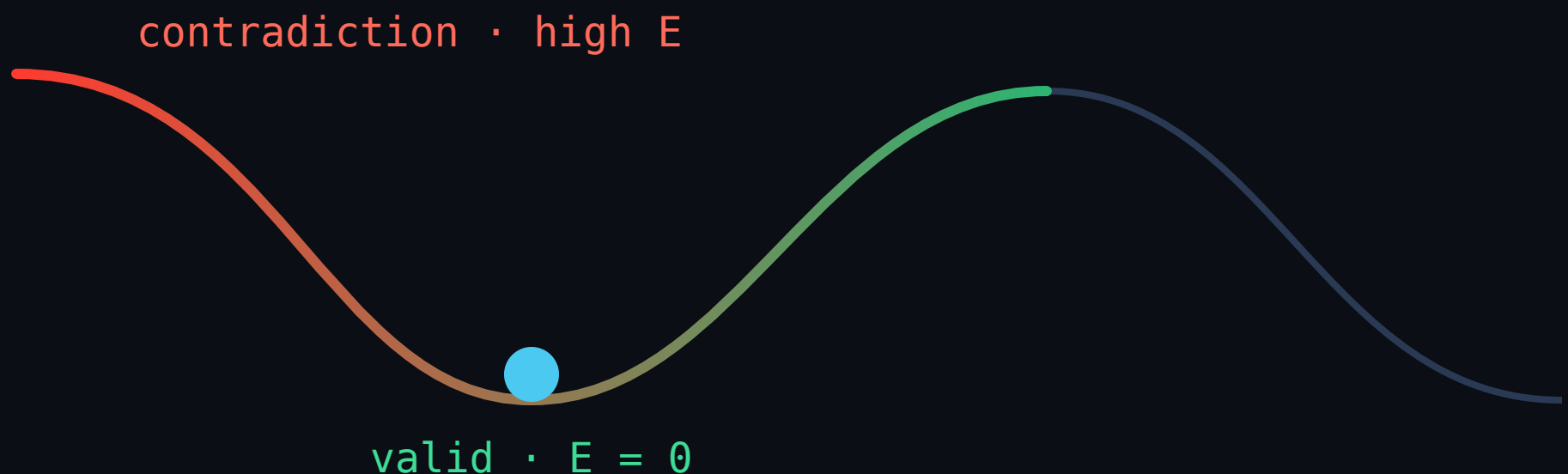
Worse: they hand it to you with **full confidence** and **no signal** that anything is wrong.

THE IDEA

Reasoning as thermodynamics

Give every answer an **energy**. Obey the rules
→ **0 energy** (a valley). Break one → **high energy** (a hill). "Thinking" is letting the answer roll **downhill** to a valley.

(An analogy for intuition — not literal physics.)



Score → Explain → Repair

1 · SCORE

Measure how much an answer **breaks your rules**.
Zero means it obeys all of them — no labels required.

2 · EXPLAIN

Surface **exactly which rules** it broke, in plain English.

3 · REPAIR

Roll it downhill to the **nearest valid answer**, changing as little as possible.

LIVE IN THE BROWSER

A working demonstrator

AI models happily produce answers that **break the rules** — an invalid setup, a self-contradiction. ThermoLogic gives every answer an **“energy” score**: obeying the rules costs zero energy, breaking one costs a lot. Then it rolls the answer **downhill to the nearest valid version** — the same idea as a ball settling into a valley. No training data needed, just your rules. Two live demos below — click around.

STEP 1 – SCORE

Measure how much an answer **breaks the rules** (its energy). Zero = all good.

STEP 2 – EXPLAIN

See **exactly which rules** it broke — in plain English.

STEP 3 – REPAIR

Auto-fix it to the **nearest valid answer**, changing as little as possible.

CONSTRAINT CONSOLE • THE PRODUCT

A pretend SaaS signup with rules like “SSO needs the Enterprise plan.”
Imagine an AI filled this in — is it valid?

TRY: pick a plan, toggle features, then hit Repair

RULE-BREAK SCORE (0 = obeys every rule)

5.96 INVALID

PLAN TIER (PICK ONE)

Free

Pro

Enterprise

FEATURES REQUESTED

SSO

Audit logs

Priority support

Custom domain

REASONING WAVE • THE FINDING

A chain of reasoning: fact A forces B, which forces C... Each bar is one link.
As the AI “thinks” longer, the truth spreads **one link at a time** — so deeper reasoning costs proportionally more thinking. A small original finding.

TRY: drag the “compute” slider and watch the wave climb

how true (0 → 1)



Toggle a configuration; the gauge scores validity live; one click repairs it to a valid state — the same engine as the PyTorch library, running client-side.

An invalid signup, fixed

✗ BEFORE — BREAKS A RULE

Pro plan · requests SSO, Audit logs, Seats>5
→ SSO needs the Enterprise plan

▼ repair (energy descent)

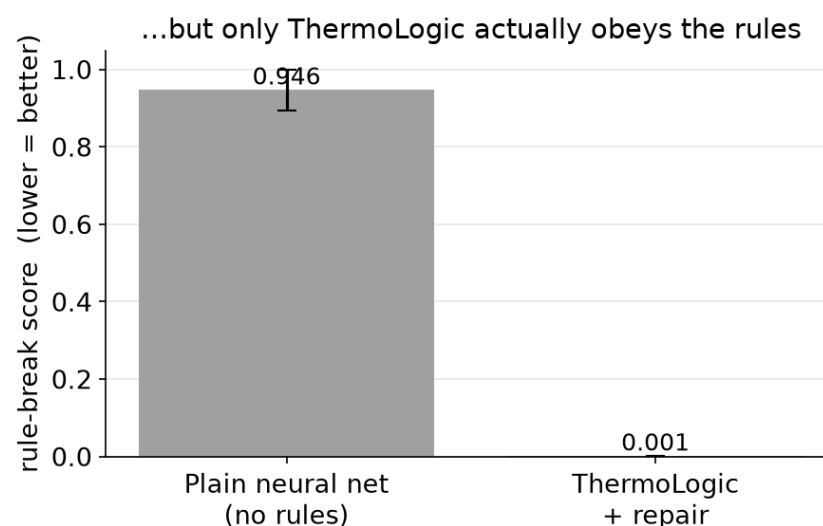
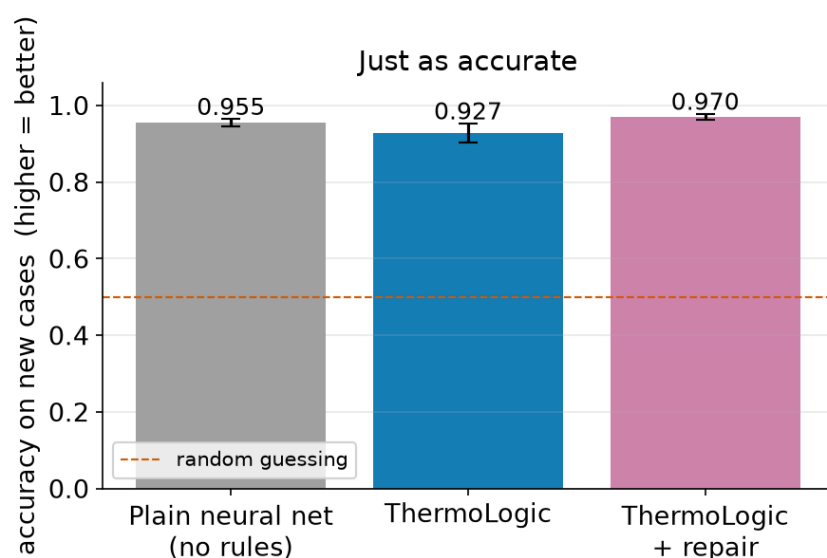
✓ AFTER — VALID

Pro plan · Audit logs, Seats>5 · SSO removed
→ obeys every rule, minimal change

FINDING 1

The value is **reliability**, not accuracy

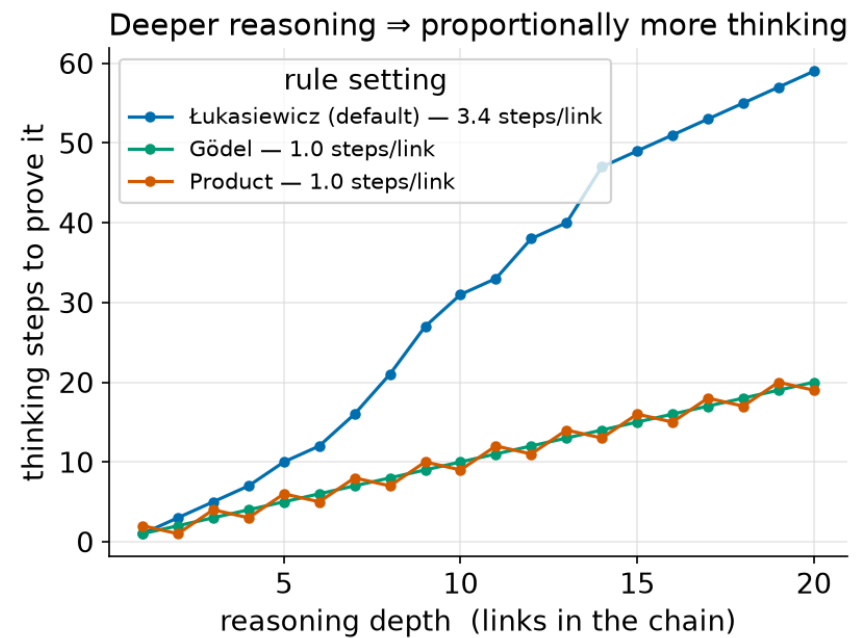
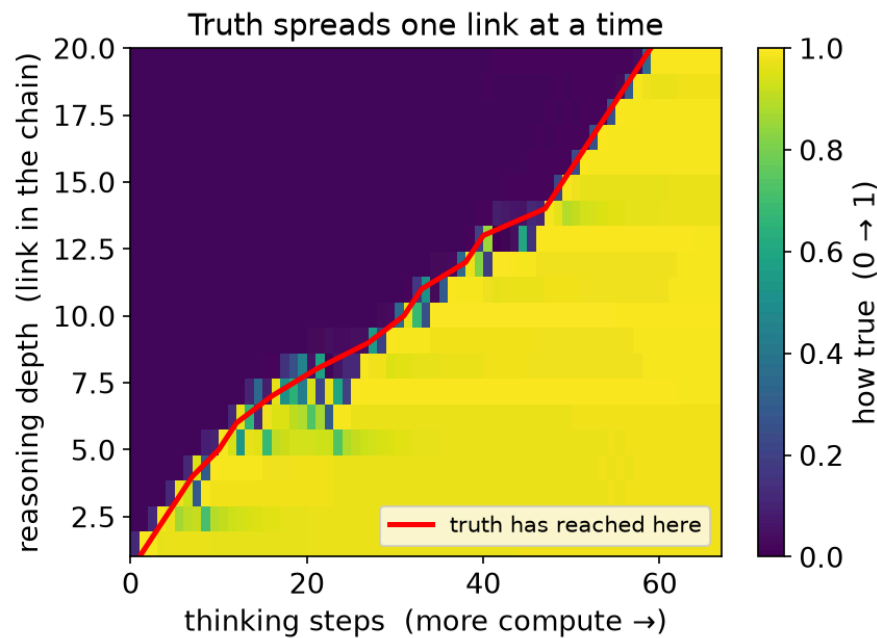
A plain net can match on accuracy — but can't tell when it's wrong



A plain neural net matches on accuracy (0.955) — but its answers are logically **inconsistent**. Only the energy layer gives a label-free **consistency guarantee** (0.001).

FINDING 2

Deeper reasoning = more compute

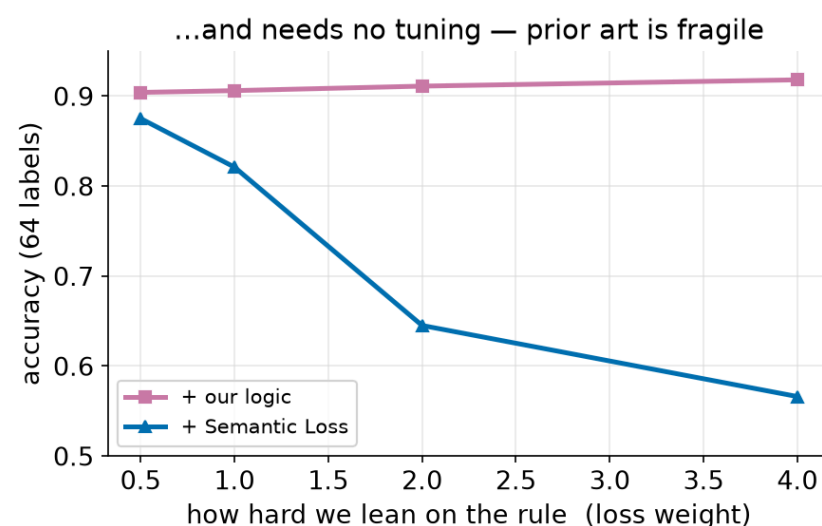
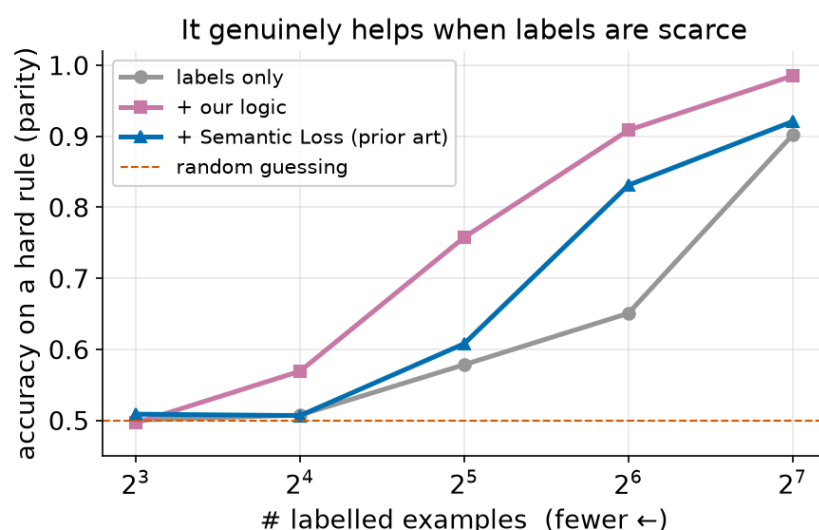


Watching the energy descent, truth propagates through a chain of rules **one hop at a time** — so twice the reasoning depth costs about twice the "thinking." A small original finding.

FINDING 3

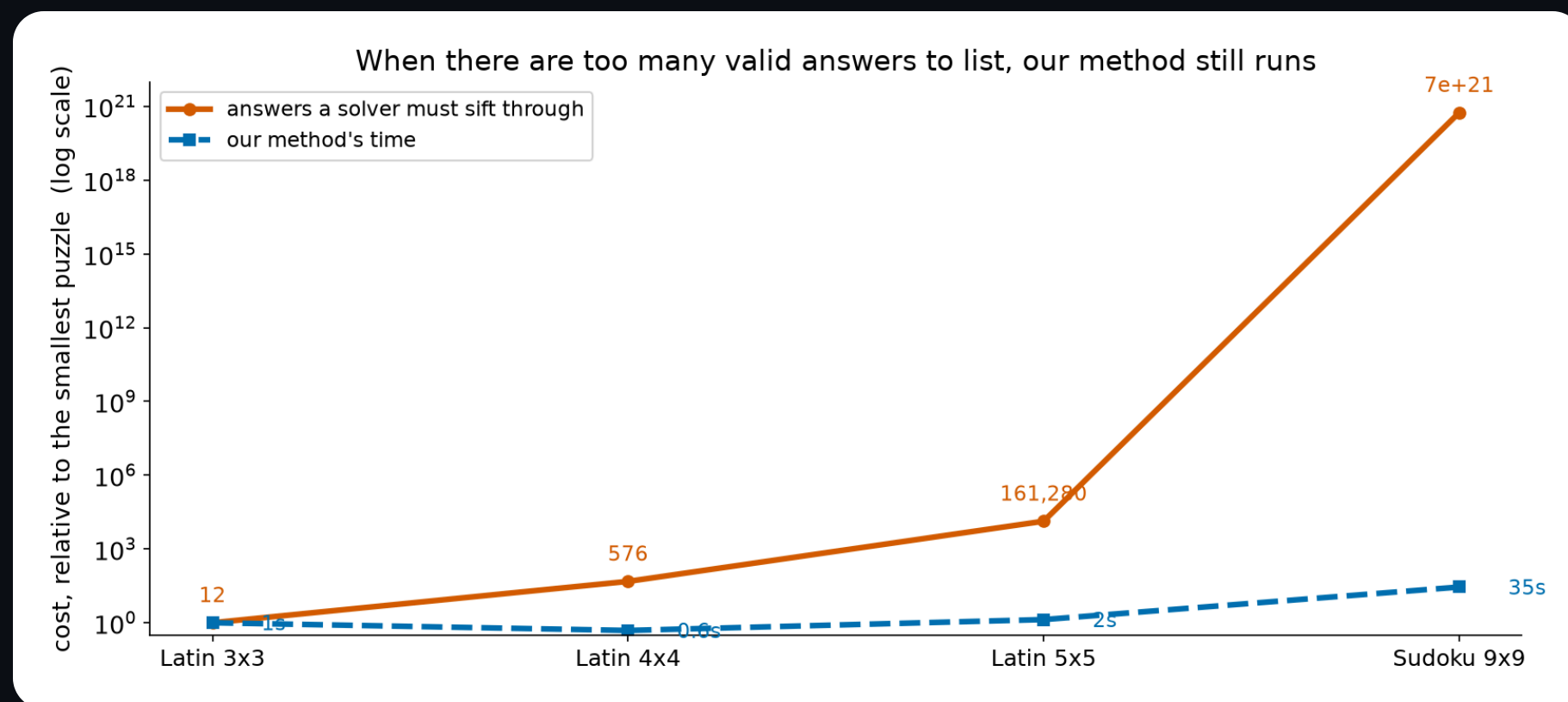
It can teach a model a **hard rule**

A distinct mechanism from Semantic Loss — as good here, and far more robust



On a rule that's hard to learn from data (parity), using the logic on **unlabelled** examples lifts accuracy by up to **+26 points** when labels are scarce — and it's a **distinct mechanism** from the closest prior art (Semantic Loss): as good here, and far more robust to tuning.

Works when there are **too many answers to list**



From a 3×3 grid up to **9×9 Sudoku** ($\approx 6.7 \times 10^{21}$ valid answers), the "list every valid answer" approach explodes ~ 20 orders of magnitude — our method keeps running in seconds. (Honest note: a purpose-built solver still handles Sudoku; the point is a **generic, trainable** layer that doesn't need one.)

I built the test designed to **make it fail**

To rebut "you graded your own homework," I ran a head-to-head on a problem I **did not design the rules for** — recovering real **Latin squares** from noise.

✗ WHERE IT LOSES

A classic solver **beats** my method here (100% vs 3.6% valid)
— for small, clean puzzles, use a solver.

✓ WHY THAT MATTERS

Reporting the loss is what makes the **wins** credible.
Reliability comes from honesty, not spin.

Where it helps – and where it doesn't

✓ **Guarantees consistency** — flags & repairs rule-breaks a plain net can't even detect

✓ **Keeps running** where "list every valid answer" explodes (measured to 9×9 Sudoku), and is **differentiable** — trainable through

✓ **+26 pts** as a semi-supervised signal on a hard rule, when labels are scarce

⚖️ **Ties** — does *not* beat an exact solver on raw accuracy

✗ **No free lunch** — with full labels and easy rules, it's a wash

What this is (and isn't)

IT IS

A reproducible **proof-of-concept** and a small usable tool. Builds openly on prior neuro-symbolic work (LTN, DeepProbLog, Semantic Loss). Every claim tied to a number.

IT ISN'T

An LLM upgrade, or a claim to have "solved" hallucination. It works today on **structured outputs with clear rules** — configs, forms, tabular decisions.

Scaling the idea toward language models is the road ahead — the vision, not a result.

TRY IT · READ IT · BUILD WITH IT

Have a look

 Live demo joperjoker.github.io/Project-ThermoLogic

 Technical report (PDF) in the repo

 Code github.com/joperjoker/Project-ThermoLogic

 `pip install -e .` from `thermologic` import `LogicEnergy`

If your team fights constraint-satisfaction on AI outputs — configs, forms, compliance — I'd love to compare notes.

— Teo Qing Cong Eugene · linkedin.com/in/eugene-teo